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Deep Learning and Its Applications (CO3133)

ASSIGNMENT-1

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1 Introduction

1.1 Background and Motivation

Image classification máp an image to a category. In an online product catalogue, for example, an deep learning classifier could help sorting images into the correct category. The classification is only one part of the system. The reliability of the result also depends on the dataset, the representation of each image, the separation between training and evaluation data and also the way model quality and computational cost are measured.

Fashion-MNIST provides a practical benchmark for investigating these factors. Its compact grayscale images allow different deep learning architectures and configurations to be evaluated efficiently, while visually similar categories introduce meaningful classification challenges.

This project investigates a range of machine learning and deep learning approaches, from simple baseline models to more advanced architectures. By comparing their predictive performance, generalization ability, training behavior, and computational requirements under a consistent experimental setup, we aim to understand the strengths, limitations, and trade-offs of different approaches to im

1.2 Problem Statement

The task is a supervised, single lable, 10 class image classifications. Let $X \in \{0, \dots, 255\}^{28\times 28}$ be a raw grayscale image and $y \in \{0, \dots, 9\}$ its label. A classifier with parameters θ receives a preprocessed representation $T(X)$ and produces ten real-valued class scores, or logits:

$$f_\theta(T(X)) = \boldsymbol{z} \in \mathbb{R}^{10}, \qquad \widehat{y} = \underset{k\in\{0,\dots,9\}}{\arg\max}\ z_k. \tag{1}$$

The task focuses on assigning a single category to each image. We use this classification problem to compare how different neural network architectures learn and represent image features.

1.3 Objectives and Scope

The objective of this project is to build an end-to-end deep learning pipeline for image classification using the Fashion-MNIST dataset. Through this project, we aim to understand how different neural network architectures process image data and how their designs influence classification performance. We implement and compare five architectures: Linear Classifier, Multilayer Perceptron (MLP), Convolutional Neural Network (CNN), LSTM, and Transformer.

The scope of this project covers the complete workflow, from data exploration and preprocessing to model training and evaluation using PyTorch. All models are trained and evaluated on the same dataset splits to ensure a fair comparison. We analyze their performance using classification accuracy, macro-F1 score, training time, inference time, and model complexity. In addition, we examine training behavior and prediction errors to better understand the strengths and limitations of each architecture.

1.4 Proposed Approach

We use a common experimental pipeline to compare the five architectures. Fashion-MNIST is divided into fixed training, validation, and test subsets, with normalization statistics computed only from the training subset. Each image is represented as a flattened vector for the Linear Classifier and MLP, a two-dimensional grid for the CNN, a sequence of rows for the LSTM, or a sequence of patches for the Transformer. Each model is trained using a shared optimization protocol, and its best checkpoint is selected by the lowest validation loss. The selected models are then evaluated on the test set to compare classification performance, computational cost, and prediction errors.

2 Dataset and Exploratory Analysis

2.1 Dataset Description

Fashion-MNIST contains pieces of fashion images divided into 10 product categories. The dataset included 60,000 train examples and 10,000 of test samples. Each image is a 28 by 28 array unsigned 8-bit gray scale intensities, the measurement range is 0 to 255. Each record has 1 interger label. The table below gives the label mapping used by the source dataset and ours implementation.

Label	**Category**	**Label**	**Category**
0	T-shirt/top	5	Sandal
1	Trouser	6	Shirt
2	Pullover	7	Sneaker
3	Dress	8	Bag
4	Coat	9	Ankle boot

Table 1: Fashion-MNIST Class Labels and Categories

The dataset includes several visually similar categories, such as T-shirt/top and Shirt, or Pullover and Coat, which makes classification more challenging. Despite the small image size, models must learn to distinguish subtle differences in shape and texture to correctly identify each category.

Fashion-MNIST is suitable for this project because its standardized image format allows different neural network architectures to be compared under consistent conditions. However, its low-resolution grayscale images do not fully represent real-world product images, which often contain colors, complex backgrounds, and greater visual variation.

2.2 Data Preprocessing

The official Fashion-MNIST training dataset is divided into training and validation subsets using a stratified 90/10 split with a fixed random seed of 42. This produces 54,000 training images and 6,000 validation images, while the official 10,000-image test set remains separate. Stratification ensures that all ten classes maintain equal representation across the partitions, as shown in Table 2.

Partition	Images	Per class	Role
Training	54,000	5,400	Fit normalization and update model parameters.
Validation	6,000	600	Select the lowest-validation-loss checkpoint.
Official test	10,000	1,000	Evaluate the fixed selected checkpoint.

Table 2: Frozen dataset partitions and their experimental roles. Counts are verified locally.

The training set is used to compute normalization statistics and update model parameters, while the validation set is used to monitor training performance and select the best checkpoint based on validation loss. The test set is reserved for the final evaluation. Figure 1 illustrates the data preparation and evaluation process.

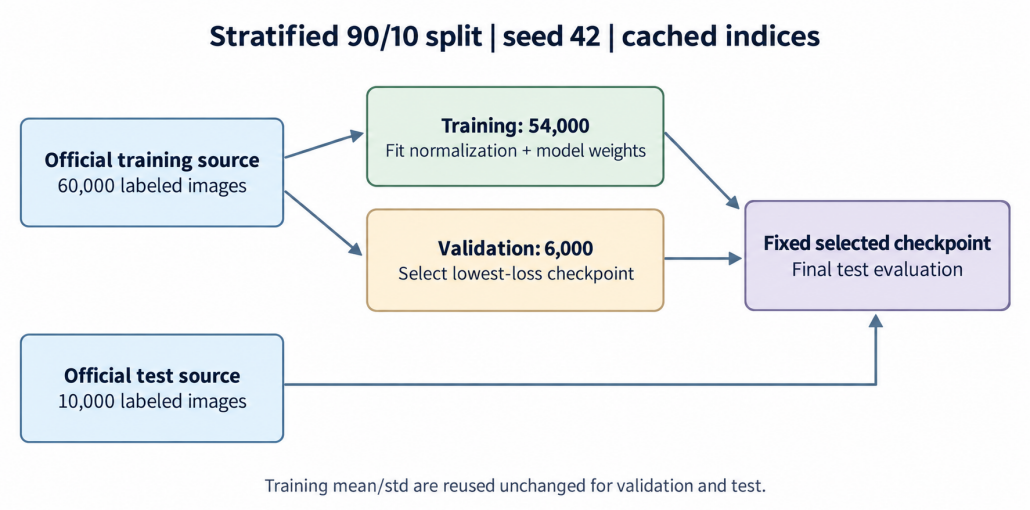


Figure 1: Data Splitting and Evaluation Pipeline

To ensure reproducibility and prevent data leakage, the training and validation indices are fixed and verified to contain no overlapping samples. An additional integrity check is performed using SHA-256 hashes to identify exact duplicate images within and across partitions. As shown in Table 3, no exact duplicates are detected. However, this method cannot identify visually similar images or different images of the same product.

Duplicate Check	Count
Duplicate images within training set	0
Duplicate images within validation set	0
Duplicate images within test set	0
Shared images between training and validation	0
Shared images between training and test	0
Shared images between validation and test	0

Table 3: Duplicate Image Analysis Across Dataset Partitions

2.3 Exploratory Data Analysis and Preprocessing

Exploratory Data Analysis (EDA) is performed on the training dataset to better understand its characteristics before model training. The analysis focuses on class distribution, image variation, pixel intensity, and similarities between categories.

2.3.1 Class Distribution.

The dataset is evenly distributed across all ten categories, with 5,400 training, 600 validation, and 1,000 test images per class, as shown in Figure 3. This balanced distribution helps prevent models from favoring certain classes due to differences in sample counts. Therefore, no additional class balancing techniques are required.

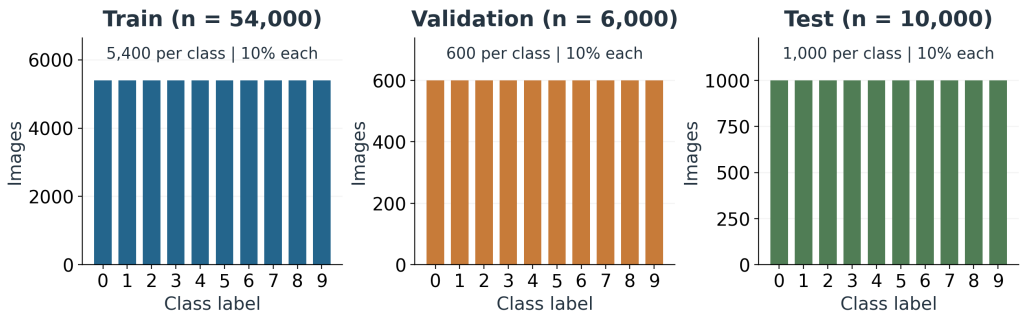


Figure 2: Class Distribution Across Dataset Partitions

2.3.2 Individual Image Variation

Figure 4 presents three randomly selected training images from each of the ten categories using a fixed random seed of 42. The examples highlight variations in shape, brightness, and appearance both within and across categories. Some categories, such as Trouser and Bag, have distinctive shapes that make them visually recognizable. In contrast, upper-body garments such as T-shirt/top, Shirt, Pullover, and Coat share similar features, making them more difficult to distinguish. These observations suggest that classification models need to learn detailed visual features rather than relying only on general shapes to accurately identify each category.

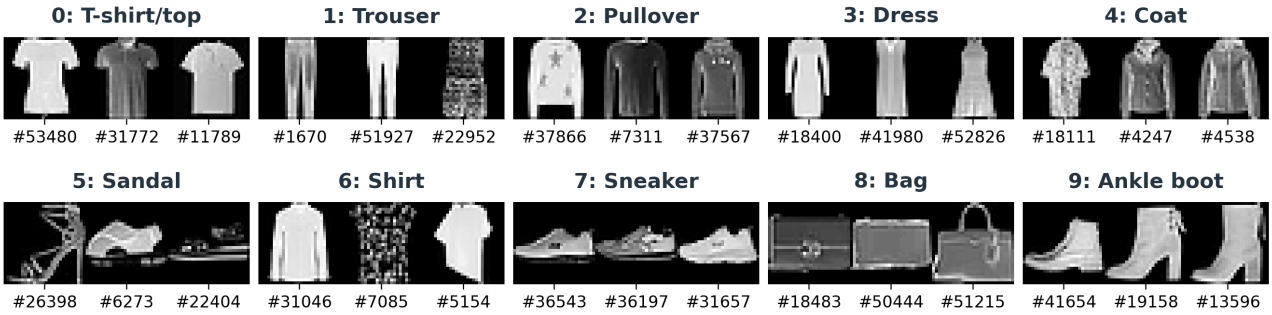


Figure 3: Sample Images from Each Fashion-MNIST Class

2.3.3 Pixel intensity and background

The training dataset contains a total of 42,336,000 pixels, with a mean intensity of 72.98 and a standard deviation of 90.06. Approximately 50.21% of the pixels have an intensity value of zero, indicating that a large portion of the images consists of dark regions. However, none of the training images is entirely blank.

Figure 5 illustrates the pixelintensity distribution. The left panel shows the distribution of all pixel values, highlighting the large proportion of zero-valued pixels. The right panel focuses on nonzero pixels to provide a clearer view of intensity variation.

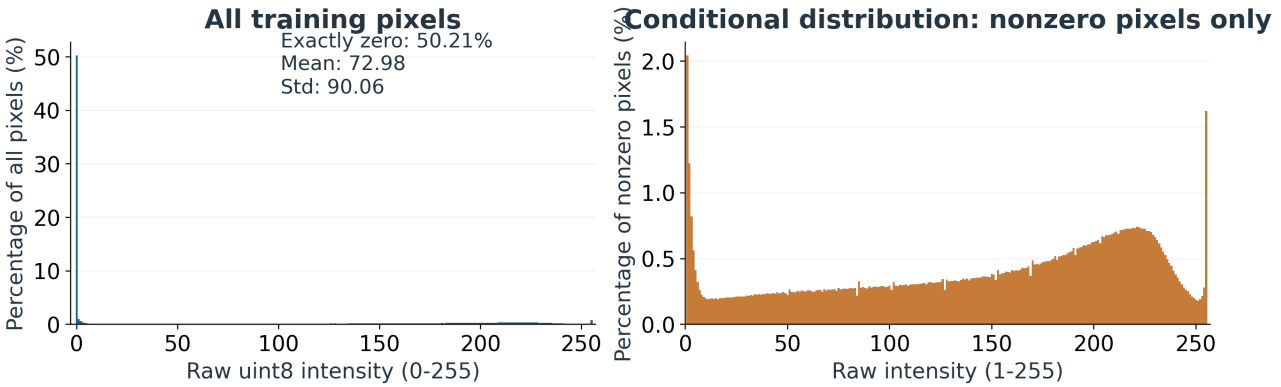


Figure 4: Training Pixel Intensity Distribution

2.3.4 Brightness and Pixel Coverage

Image brightness is calculated as the average intensity of all 784 pixels, ranging from 0 to 255. Nonzero pixel coverage represents the percentage of pixels with intensity greater than zero, providing an approximate measure of how much of an image contains visible information.

Figure 5 illustrates the distributions of brightness and pixel coverage across the ten categories, while Table 4 summarizes their statistics. Coat has the highest mean brightness (98.15), whereas Sandal has the lowest (34.87). In terms of pixel coverage, Pullover has the highest average (64.84%), while Sandal has the lowest (32.10%).

Although these characteristics differ across categories, their distributions overlap considerably. This suggests that brightness and pixel coverage alone are insufficient to distinguish all classes, highlighting the importance of learning more detailed visual features for accurate classification.

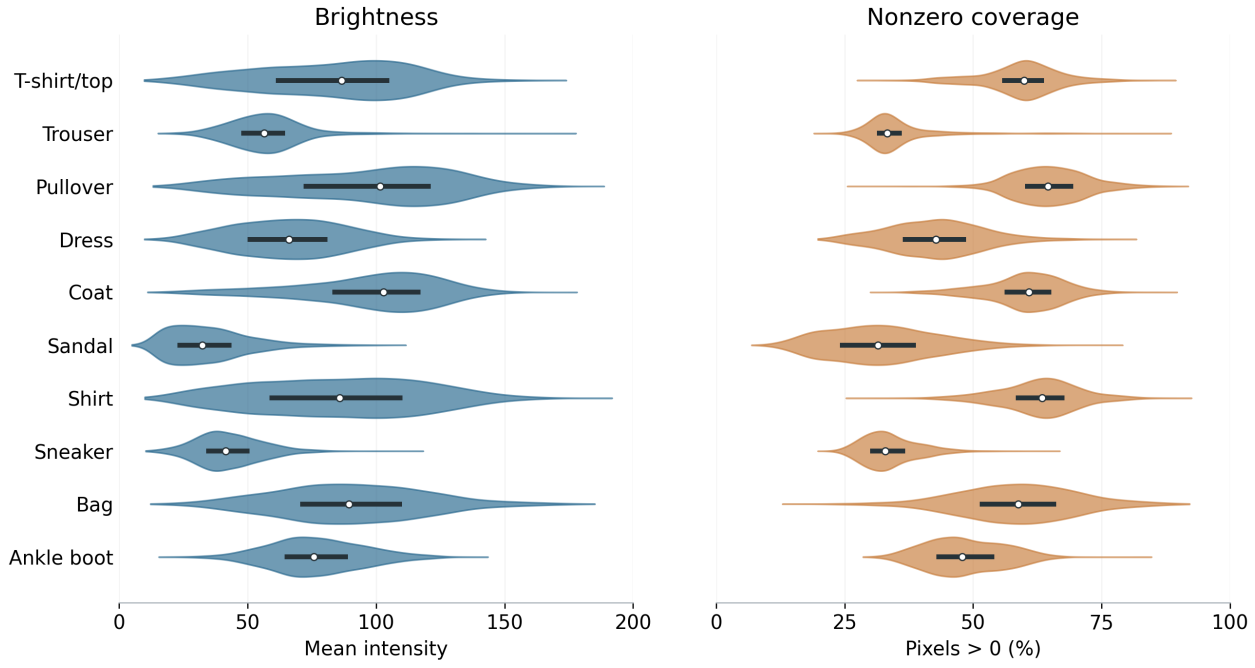


Figure 5: Distribution of Image Brightness and Nonzero Pixel Coverage by Class

Label	Class	Mean Brightness	Brightness SD	Mean Nonzero (%)
0	T-shirt/top	83.02	28.84	59.38
1	Trouser	56.91	14.55	34.91
2	Pullover	96.17	32.70	64.84
3	Dress	65.91	21.33	42.67
4	Coat	98.15	26.46	60.19
5	Sandal	34.87	15.74	32.10
6	Shirt	84.84	32.90	62.87
7	Sneaker	42.81	12.85	33.81
8	Bag	90.28	28.81	58.61
9	Ankle boot	76.87	18.32	48.48

Table 4: Image Brightness and Nonzero Pixel Statistics by Class

2.3.5 Spatial Class Averages and Variability

To examine the spatial characteristics of each class, we compute the mean image and pixel-wise standard deviation from its 5,400 training samples. The mean image represents the typical class appearance, while the standard deviation indicates variations in pixel intensity.

For each class k, these statistics are defined as:

$$\begin{aligned} M_{k,r,c} &= \frac{1}{n_k} \sum_{i \in I_k} U_{i,r,c}, \\ S_{k,r,c} &= \sqrt{\frac{1}{n_k} \sum_{i \in I_k} (U_{i,r,c} - M_{k,r,c})^2}. \end{aligned} \tag{2}$$

where $U_i = X_i/255$ is the normalized image, I_k is the set of samples in class k, $n_k = 5{,}400$, and (r, c) denotes a pixel position.

Figure 6 shows that mean images preserve recognizable class shapes, although some categories, such as pullovers and coats, appear similar. Higher standard deviations occur mainly around object boundaries, reflecting differences in shape and positioning. These patterns highlight the challenges of within-class variation and visual similarity between categories.

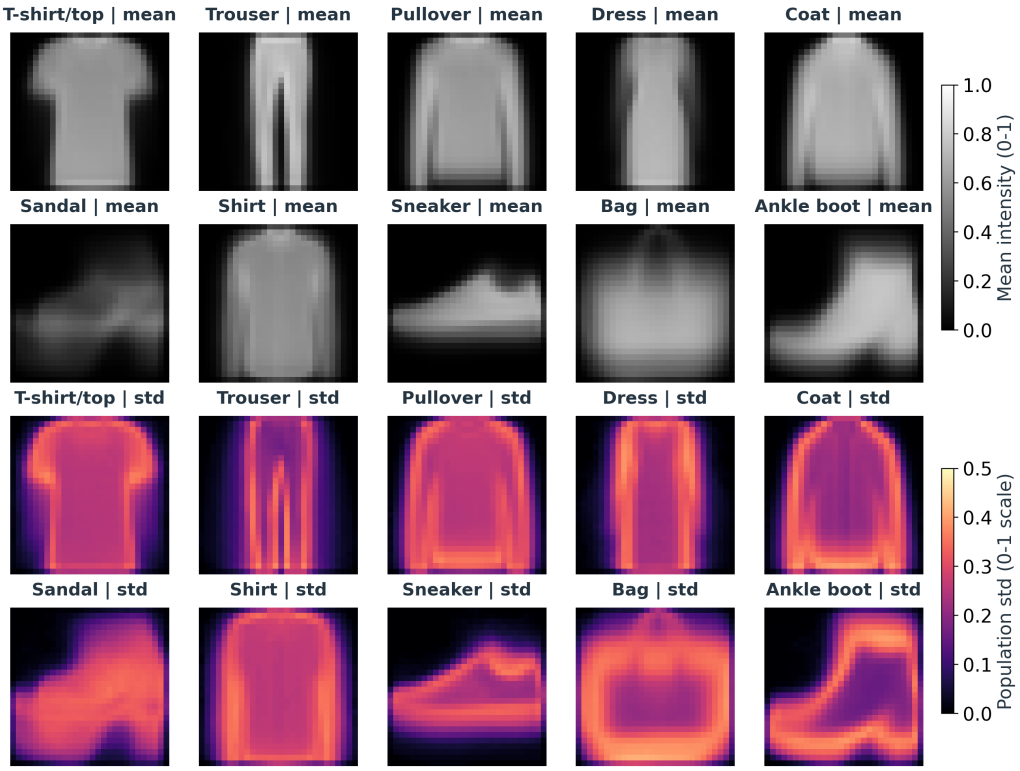


Figure 6: Class-Wise Mean Images and Pixel Variability

2.3.6 Similarity Between Class Prototypes

To quantify similarities between clothing categories, each class mean image is flattened into a 784-dimensional vector. Pairwise cosine similarity is then calculated as:

$$s_{k\ell} = \frac{\mathbf{m}_k^\top \mathbf{m}_\ell}{\|\mathbf{m}_k\|_2 \|\mathbf{m}_\ell\|_2}, \qquad \mathbf{m}_k = \mathrm{vec}(M_k) \in \mathbb{R}^{784}. \tag{3}$$

Figure 7 presents the resulting similarity matrix. The highest similarity between distinct classes is 0.9873, observed between Coat and Shirt, indicating that their average images share highly similar spatial patterns.

These results highlight potential challenges in distinguishing visually similar categories. However, cosine similarity only measures resemblance between class averages and does not account for individual image variations or model predictions. Therefore, the observed similarities will be compared with model confusion matrices during the evaluation stage.

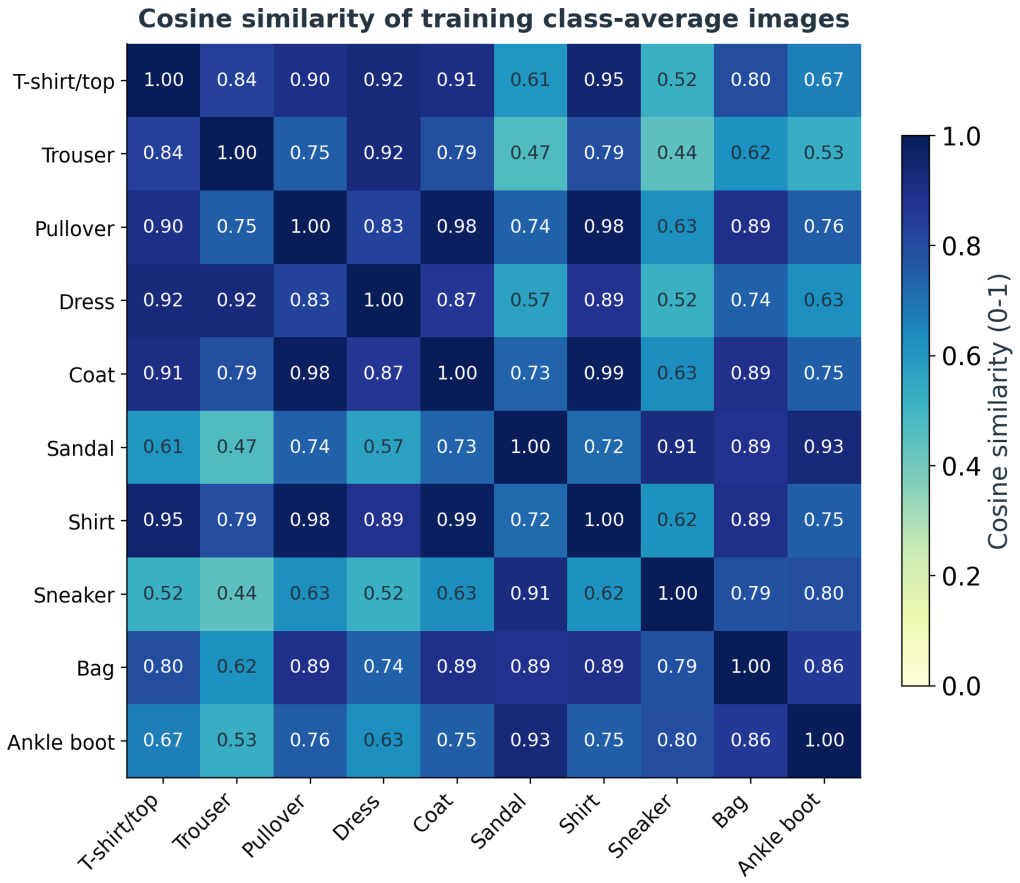


Figure 7: Cosine similarity between training class-average images.

2.4 Preprocessing